COGNIZANT DIGITAL NURTURE 5.0

PL/SQL EXERCISE 1 – CONTROL STRUCTURES

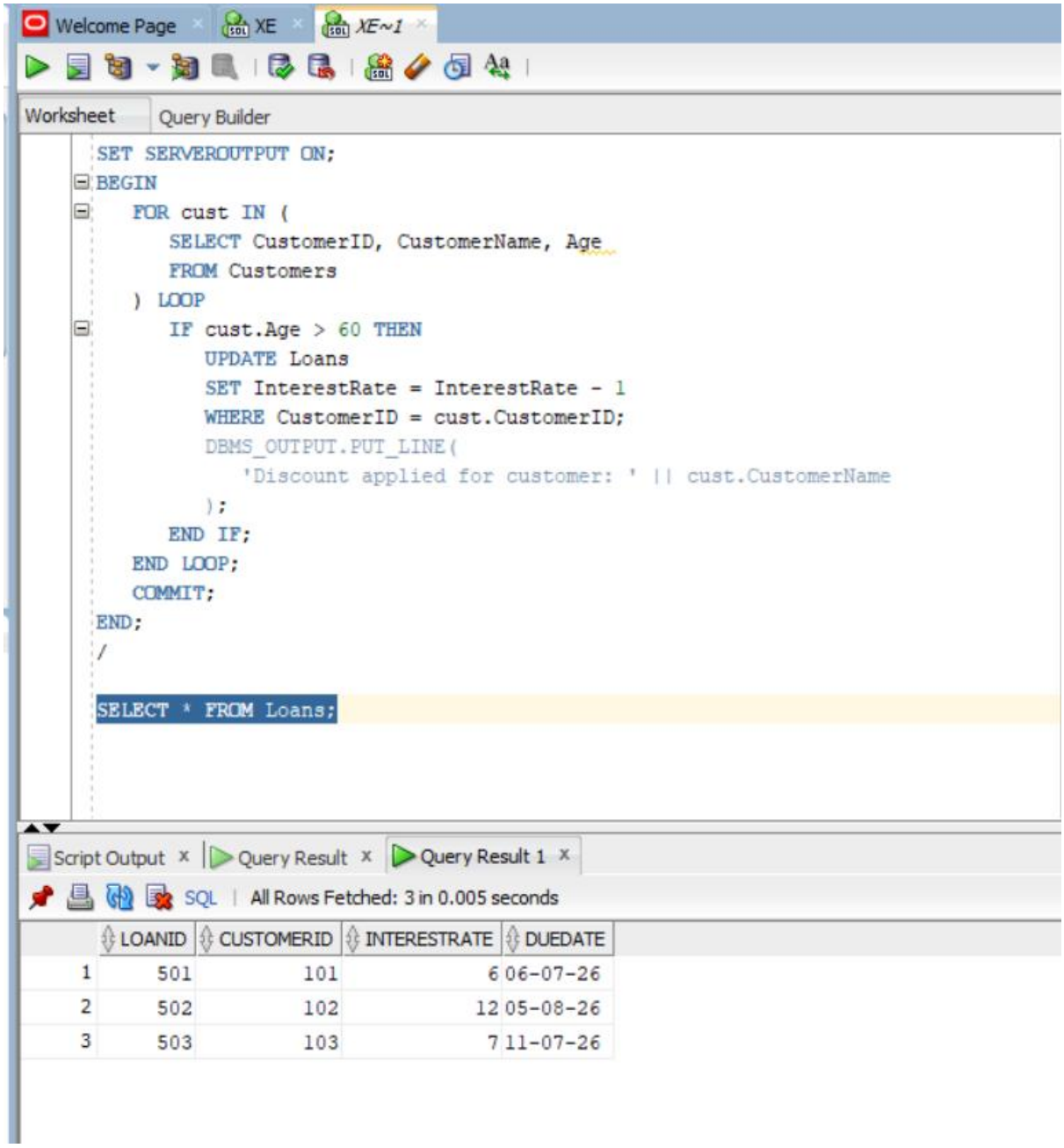
SCENARIO 1: APPLY DISCOUNT TO LOAN INTEREST RATES

Problem Statement:
The bank wants to apply a 1% discount to loan interest rates for customers above 60 years old.

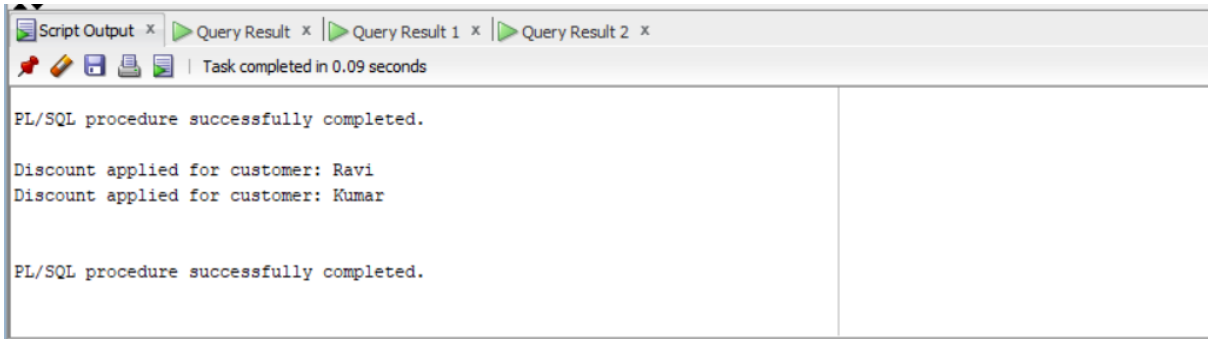
PL/SQL Code:

```
SET SERVEROUTPUT ON;

BEGIN
FOR cust IN (
SELECT CustomerID, CustomerName, Age
FROM Customers
) LOOP

  IF cust.Age > 60 THEN

    UPDATE Loans

    SET InterestRate = InterestRate - 1

    WHERE CustomerID = cust.CustomerID;


    DBMS_OUTPUT.PUT_LINE(

      'Discount applied for customer: ' || cust.CustomerName

    );

  END IF;

END LOOP;

COMMIT;
END;
/
```

Output Screenshot:



EXECUTION OUTPUT:



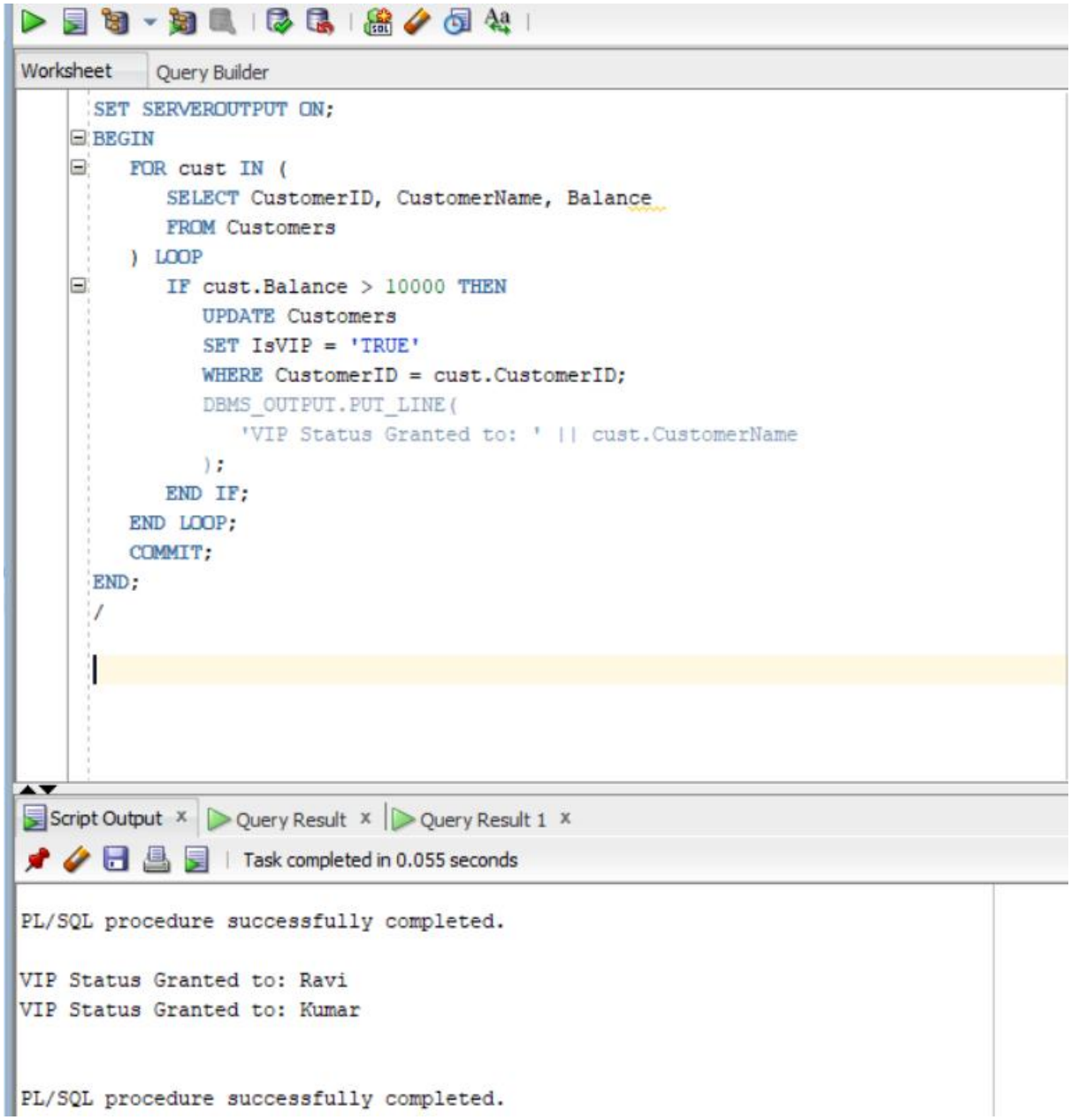
SCENARIO 2: PROMOTE CUSTOMERS TO VIP STATUS

Problem Statement:
A customer can be promoted to VIP status based on their balance.

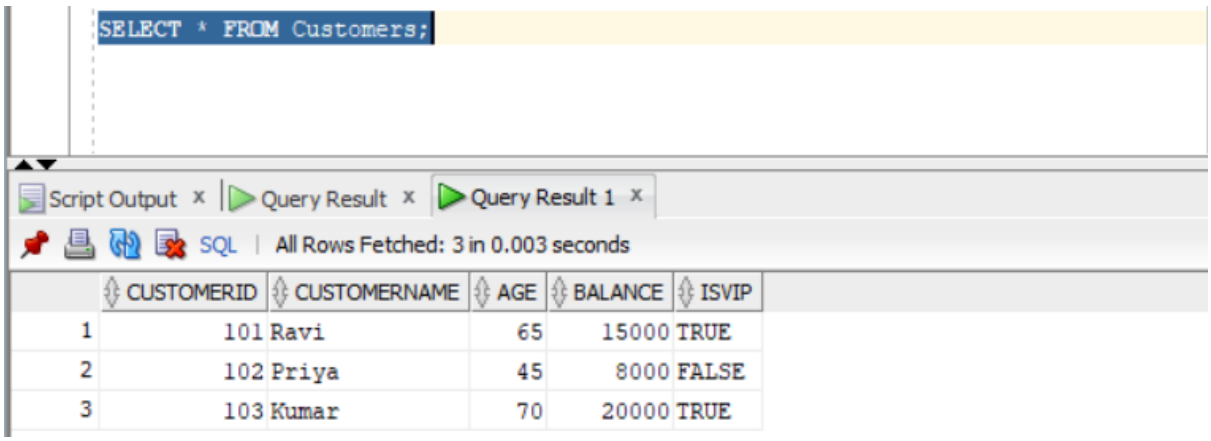
PL/SQL Code:

```
SET SERVEROUTPUT ON;

BEGIN
FOR cust IN (
SELECT CustomerID, CustomerName, Balance
FROM Customers
) LOOP

  IF cust.Balance > 10000 THEN

    UPDATE Customers

    SET IsVIP = 'TRUE'

    WHERE CustomerID = cust.CustomerID;


    DBMS_OUTPUT.PUT_LINE(

      'VIP Status Granted to: ' || cust.CustomerName

    );

  END IF;

END LOOP;

COMMIT;
END;
/
```

Output Screenshot:



FINAL OUTPUT



	CUSTOMERID	CUSTOMERNAME	AGE	BALANCE	ISVIP
1	101	Ravi	65	15000	TRUE
2	102	Priya	45	8000	FALSE
3	103	Kumar	70	20000	TRUE

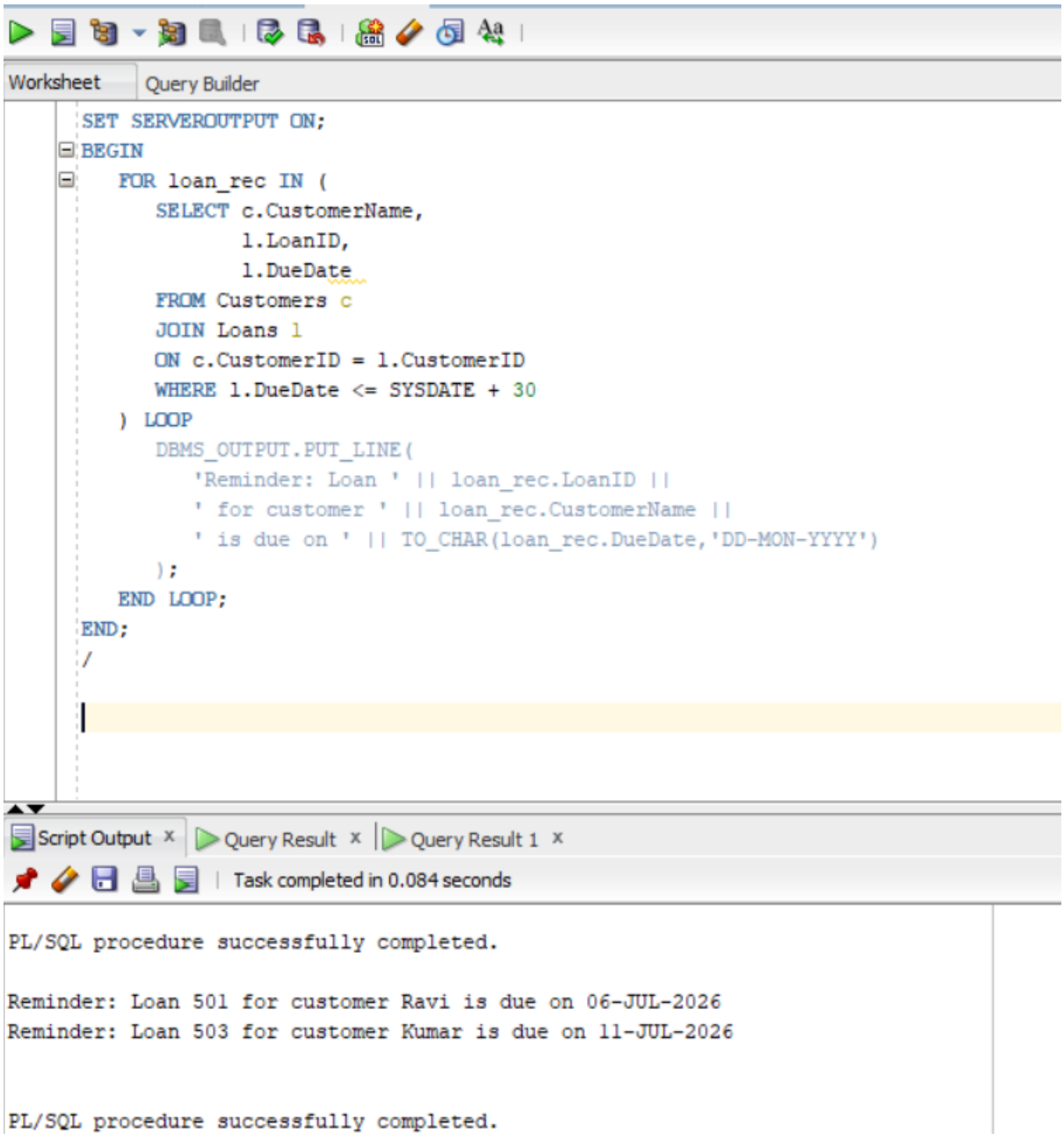
SCENARIO 3: LOAN DUE REMINDERS

Problem Statement:
The bank wants to send reminders to customers whose loans are due within the next 30 days.

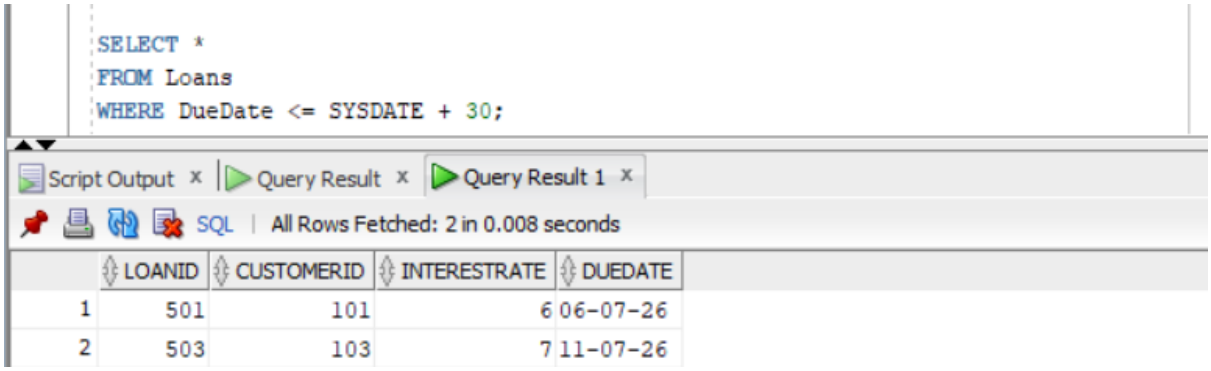
PL/SQL Code:

```
SET SERVEROUTPUT ON;

BEGIN
FOR loan_rec IN (
SELECT c.CustomerName,
l.LoanID,
l.DueDate
FROM Customers c
JOIN Loans l
ON c.CustomerID = l.CustomerID
WHERE l.DueDate <= SYSDATE + 30
) LOOP

  DBMS_OUTPUT.PUT_LINE(

    'Reminder: Loan ' || loan_rec.LoanID ||

    ' for customer ' || loan_rec.CustomerName ||

    ' is due on ' || TO_CHAR(loan_rec.DueDate,'DD-MON-YYYY')

  );

END LOOP;
END;
/
```

Output Screenshot:

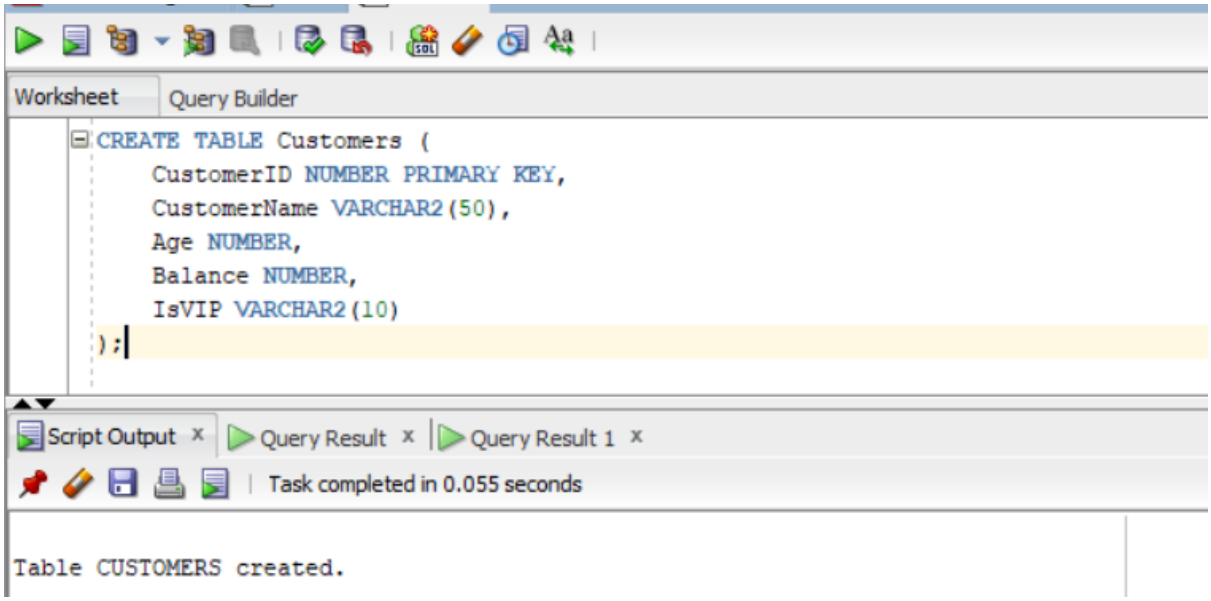


FINALOUTPUT:

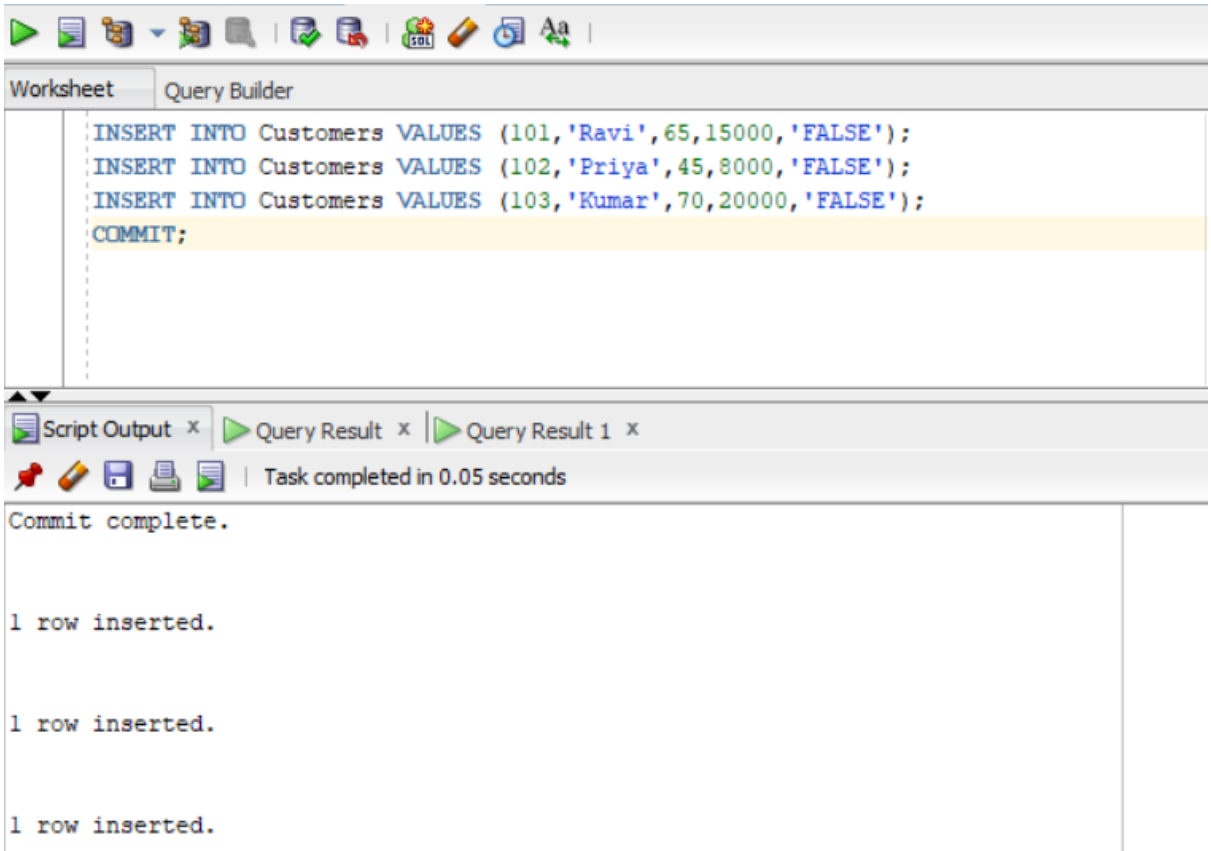


DATABASE_SETUP:

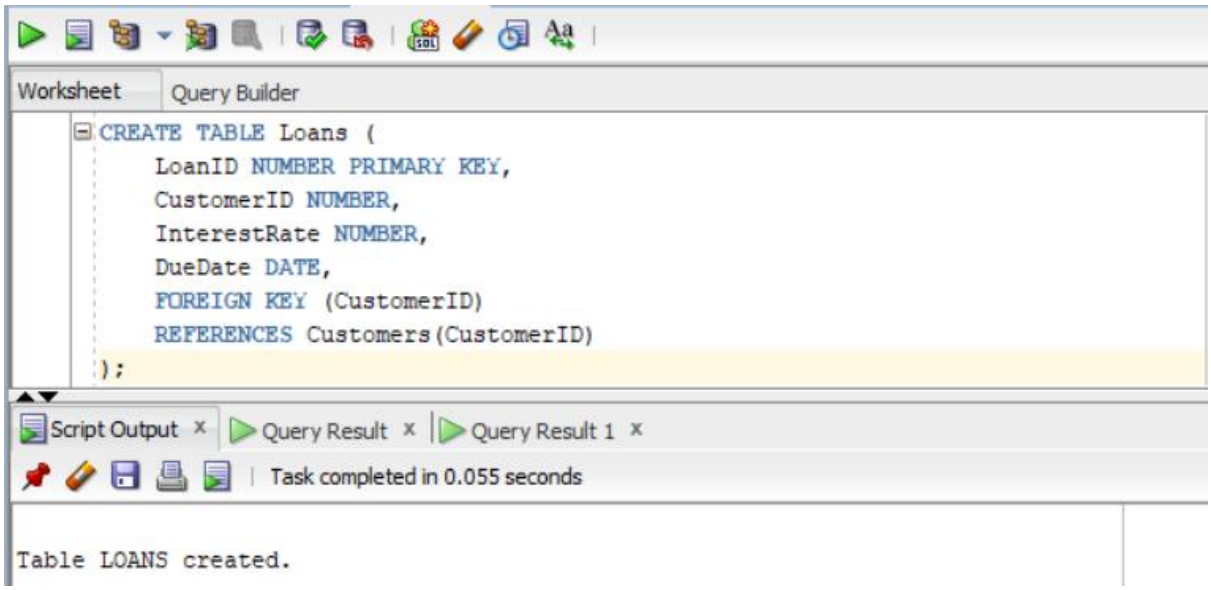
CUSTOMER TABLE CREATION:



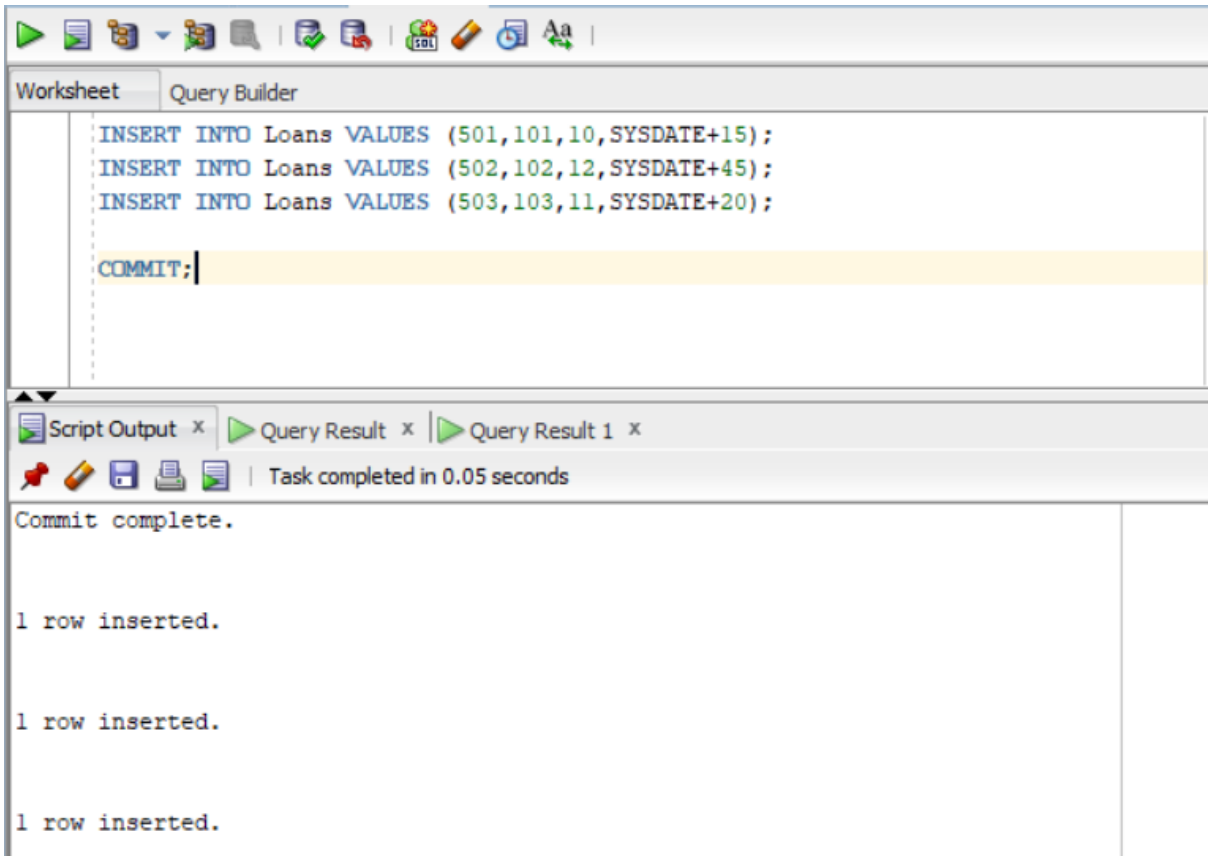
INSERTING ROWS:



LOAN TABLE CREATION:



INSERTING ROWS:



RESULT:

Thus, the PL/SQL programs using control structures were successfully implemented and executed.